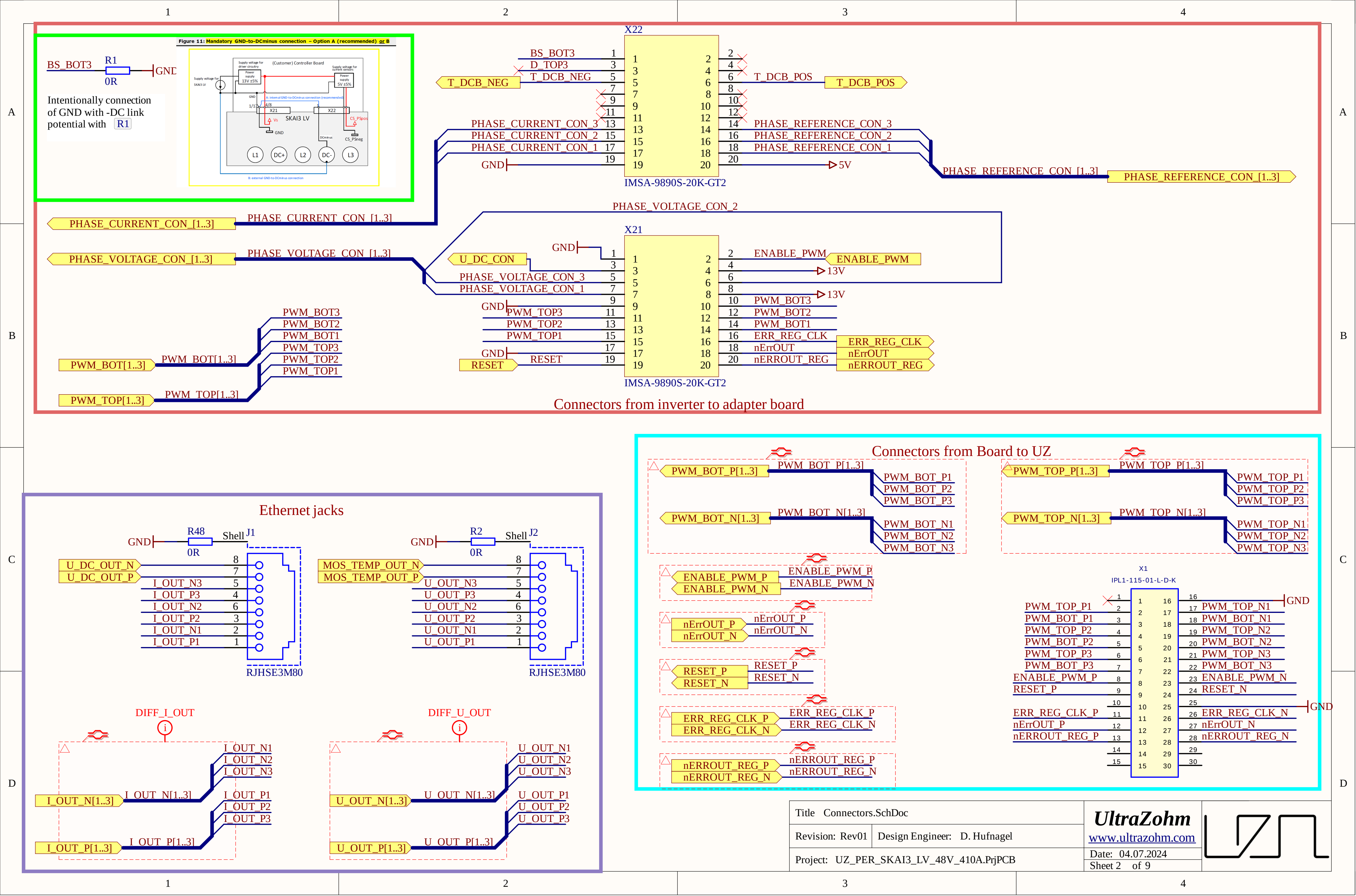
Title TopSheet.SchDoc		**UltraZohm** www.ultrazohm.com
Revision: Rev01	Design Engineer: D. Hufnagel	
Project: UZ_PER_SKAI3_LV_48V_410A.PrjPCB		Date: 04.07.2024 Sheet 1 of 9



Connectors from inverter to adapter board
BS_BOT3
R1
0R
GND
Intentionally connection of GND with -DC link potential with R1
Figure 11: Mandatory GND-to-DCminus connection – Option A (recommended) or B
(Customer) Controller Board
SKAI3 LV
X22
IMSA-9890S-20K-GT2
BS_BOT3
D_TOP3
T_DCB_NEG
T_DCB_POS
PHASE_CURRENT_CON_3
PHASE_CURRENT_CON_2
PHASE_CURRENT_CON_1
PHASE_REFERENCE_CON_3
PHASE_REFERENCE_CON_2
PHASE_REFERENCE_CON_1
PHASE_REFERENCE_CON_[1..3]
5V
PHASE_CURRENT_CON_[1..3]
PHASE_VOLTAGE_CON_[1..3]
PHASE_VOLTAGE_CON_2
X21
IMSA-9890S-20K-GT2
U_DC_CON
PHASE_VOLTAGE_CON_3
PHASE_VOLTAGE_CON_1
ENABLE_PWM
13V
PWM_TOP3
PWM_TOP2
PWM_TOP1
PWM_BOT3
PWM_BOT2
PWM_BOT1
ERR_REG_CLK
nErrOUT
nERROUT_REG
RESET
PWM_BOT[1..3]
PWM_TOP[1..3]
Connectors from Board to UZ
PWM_BOT_P[1..3]
PWM_BOT_P1
PWM_BOT_P2
PWM_BOT_P3
PWM_BOT_N[1..3]
PWM_BOT_N1
PWM_BOT_N2
PWM_BOT_N3
PWM_TOP_P[1..3]
PWM_TOP_P1
PWM_TOP_P2
PWM_TOP_P3
PWM_TOP_N[1..3]
PWM_TOP_N1
PWM_TOP_N2
PWM_TOP_N3
ENABLE_PWM_P
ENABLE_PWM_N
nErrOUT_P
nErrOUT_N
RESET_P
RESET_N
ERR_REG_CLK_P
ERR_REG_CLK_N
nERROUT_REG_P
nERROUT_REG_N
X1
IPL1-115-01-L-D-K
Ethernet jacks
R48
0R
Shell
J1
U_DC_OUT_N
U_DC_OUT_P
I_OUT_N3
I_OUT_P3
I_OUT_N2
I_OUT_P2
I_OUT_N1
I_OUT_P1
RJHSE3M80
R2
0R
J2
MOS_TEMP_OUT_N
MOS_TEMP_OUT_P
U_OUT_N3
U_OUT_P3
U_OUT_N2
U_OUT_P2
U_OUT_N1
U_OUT_P1
RJHSE3M80
DIFF_I_OUT
I_OUT_N[1..3]
I_OUT_P[1..3]
DIFF_U_OUT
U_OUT_N[1..3]
U_OUT_P[1..3]
Title Connectors.SchDoc
Revision: Rev01
Design Engineer: D. Hufnagel
Project: UZ_PER_SKAI3_LV_48V_410A.PrjPCB
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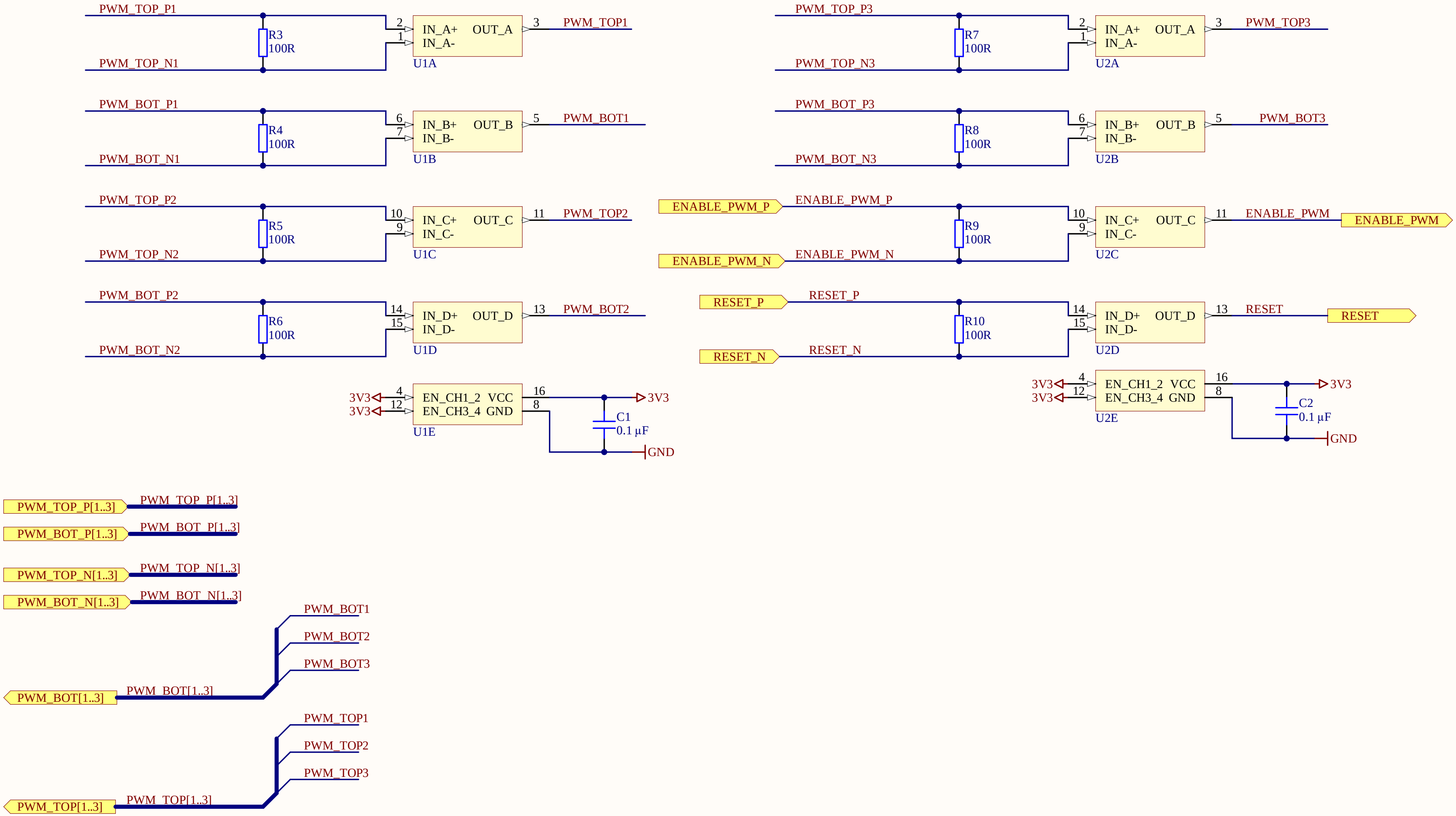
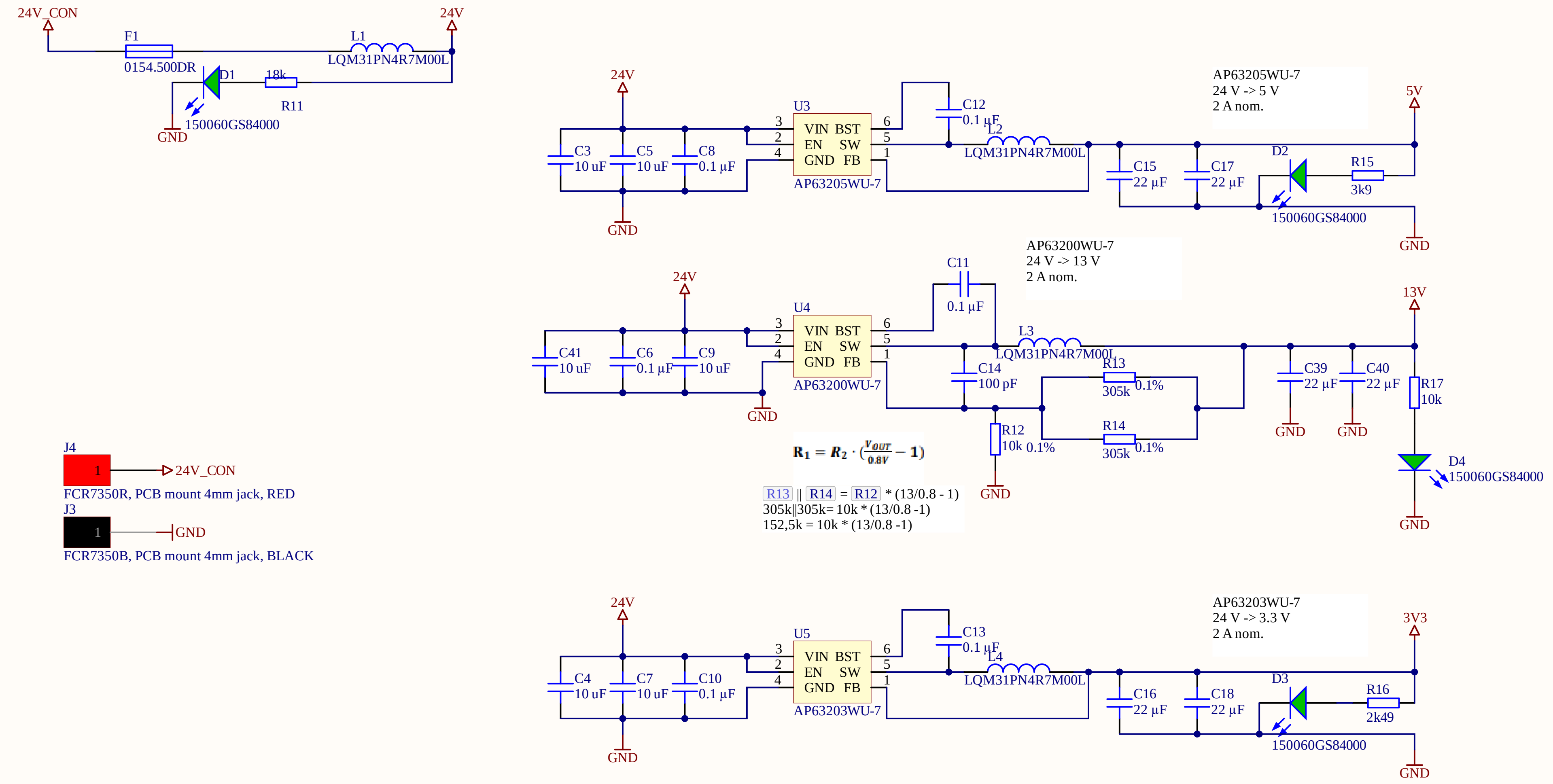
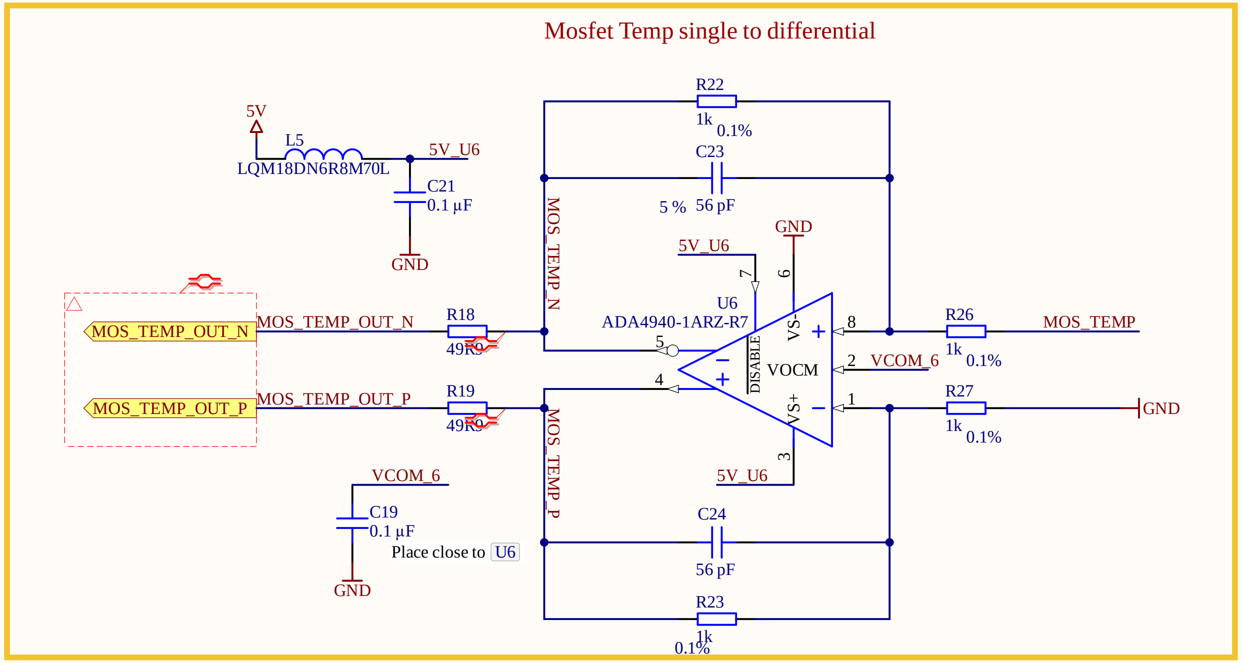
Date: 04.07.2024
Sheet 2 of 9

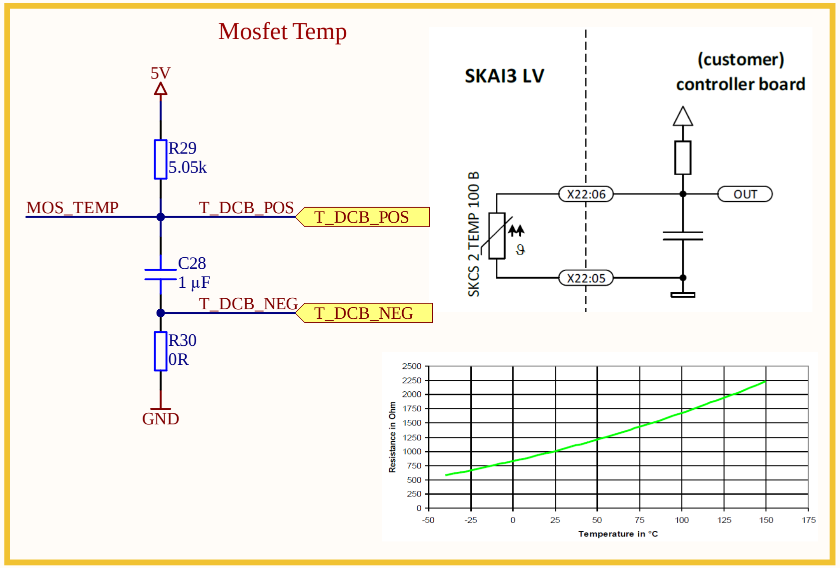


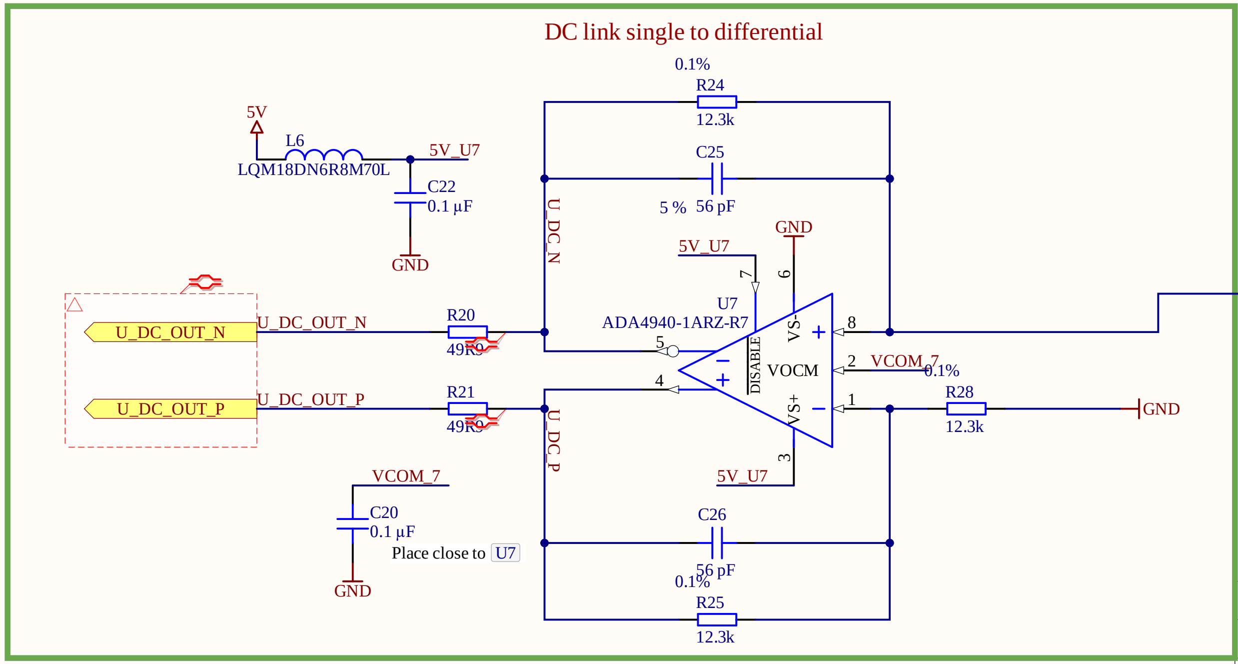
Title Receiver.SchDoc	**UltraZohm**
Revision: Rev01 · Design Engineer: D. Hufnagel	www.ultrazohm.com
Project: UZ_PER_SKAI3_LV_48V_410A.PrjPCB	Date: 04.07.2024 · Sheet 3 of 9



Title Power.SchDoc		UltraZohm www.ultrazohm.com
Revision: Rev01	Design Engineer: D. Hufnagel	
Project: UZ_PER_SKAI3_LV_48V_410A.PrjPCB		Date: 04.07.2024 Sheet 4 of 9







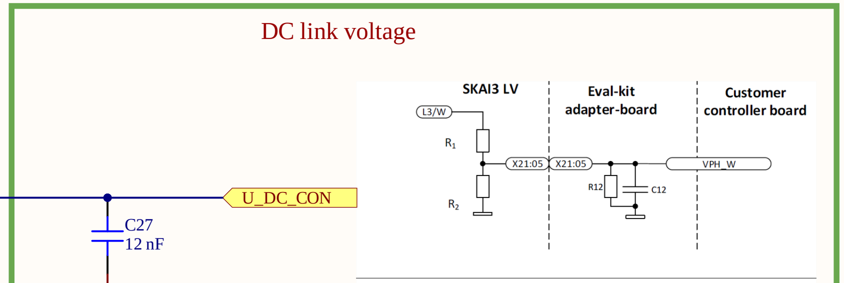
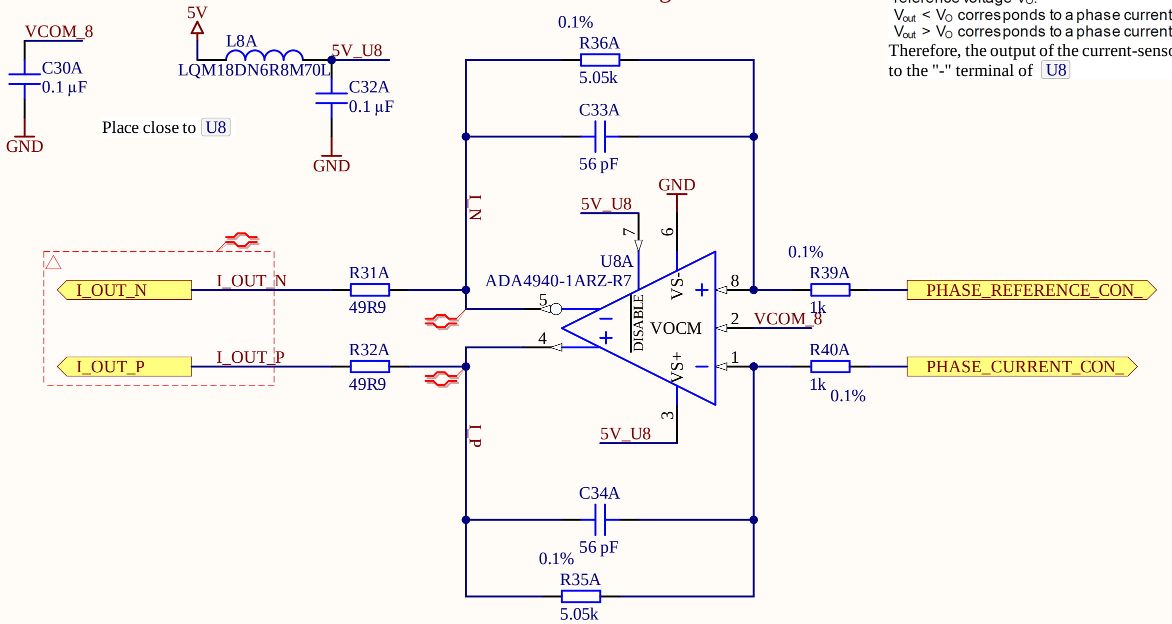


Table 3: Voltage divider resistor values

SKAI3 LV voltage class	R_1	R_2
SKAI XX A3 MD10-X	225 kΩ ± 1%	13.0 kΩ ± 1%
SKAI XX A3 MD15-X	330 kΩ ± 1%	13.0 kΩ ± 1%

Title Temp_V_DC_Measurement.SchDoc	**UltraZohm** www.ultrazohm.com
Revision: Rev01 Design Engineer: D. Hufnagel	
Project: UZ_PER_SKAI3_LV_48V_410A.PrjPCB	Date: 04.07.2024 Sheet 5 of 9

Phase-Current measurement single to differential



Please note the inverted polarity sense between primary current I_P and the difference of sensor output voltage V_{out} and reference voltage V_O:

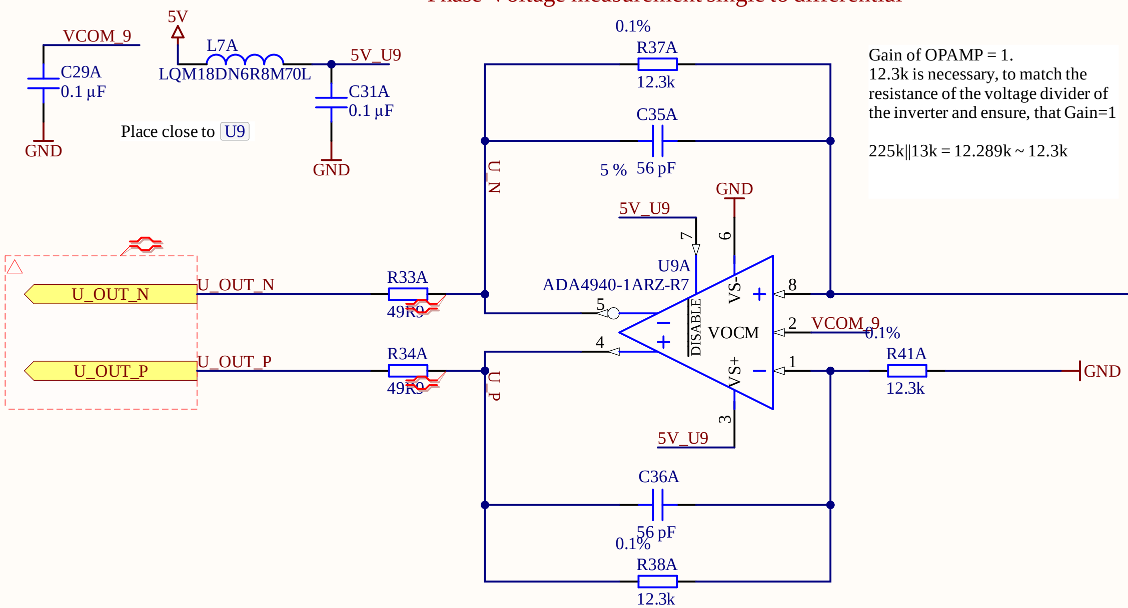
$V_{out} < V_O$ corresponds to a phase current flowing out of the SKAI3 LV

$V_{out} > V_O$ corresponds to a phase current flowing into the SKAI3 LV

Therefore, the output of the current-sensors of the inverter are connected to the "-" terminal of U8

LEM on Inverter has ±900A with V_ref=2.5V.
and 2.22mV/A
Maximum measured current will be 400A
-> 400A * ±2.22mV/a = ±0.888V in respect to V_ref.
To achieve full 5V for the ADCs, gain is set to:
-> ±0.888V * 5.05 = ±4.48V
-> 1k R39 R40
-> 5.05k R35 R36

Phase-Voltage measurement single to differential

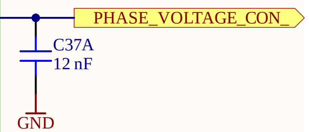


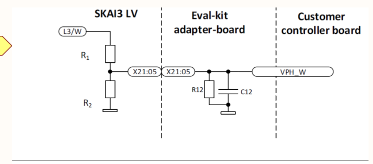
Gain of OPAMP = 1.
12.3k is necessary, to match the resistance of the voltage divider of the inverter and ensure, that Gain=1

225k||13k = 12.289k ~ 12.3k

Phase voltages

Voltage divider on the inverter with 225k and 13k @48V results in 2.62V





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Title Curr_Volt_Measurement.SchDoc

Revision: Rev01 Design Engineer: D. Hufnagel

Project: UZ_PER_SKAI3_LV_48V_410A.PrjPCB

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Date: 04.07.2024

Phase-Current measurement single to differential

Please note the inverted polarity sense between primary current I_P and the difference of sensor output voltage V_{out} and reference voltage V_O:
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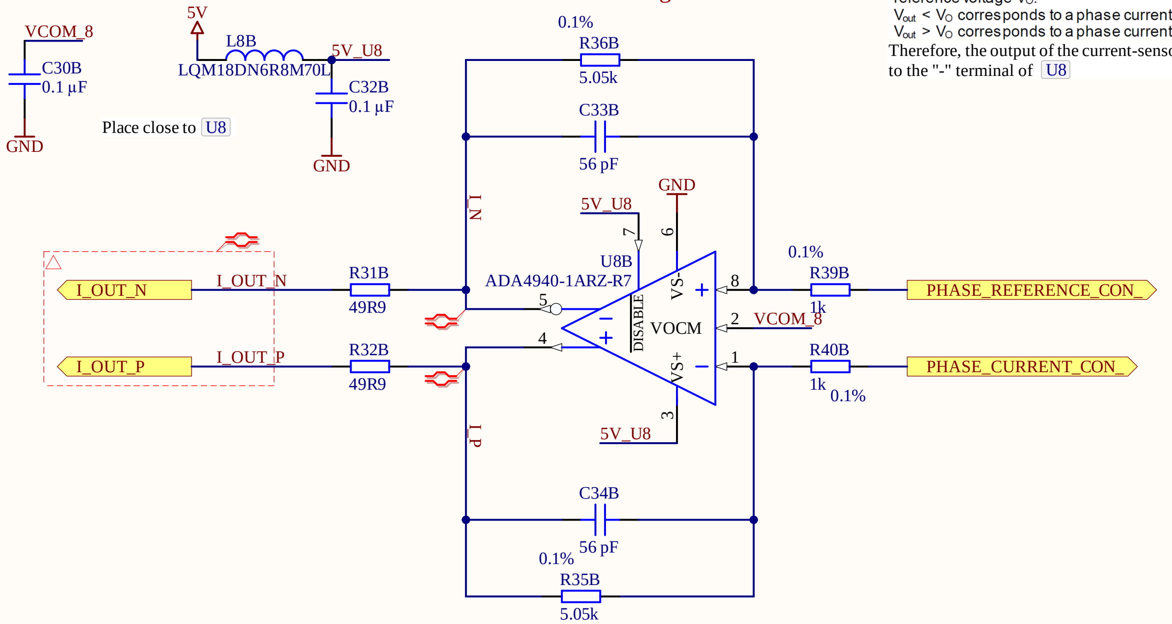

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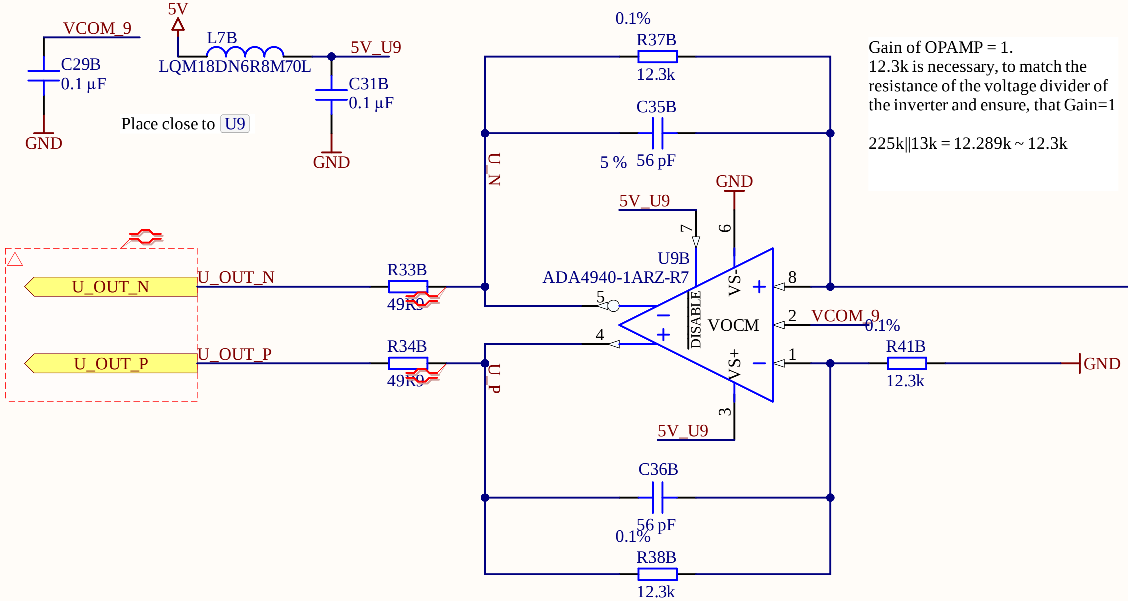
Phase-Voltage measurement single to differential



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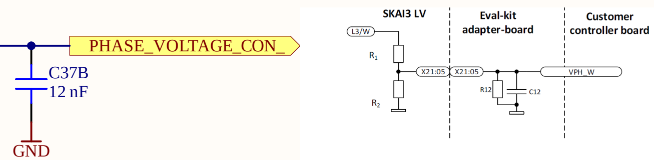
Phase voltages

Voltage divider on the inverter with 225k and 13k @48V results in 2.62V



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Title Curr_Volt_Measurement.SchDoc		
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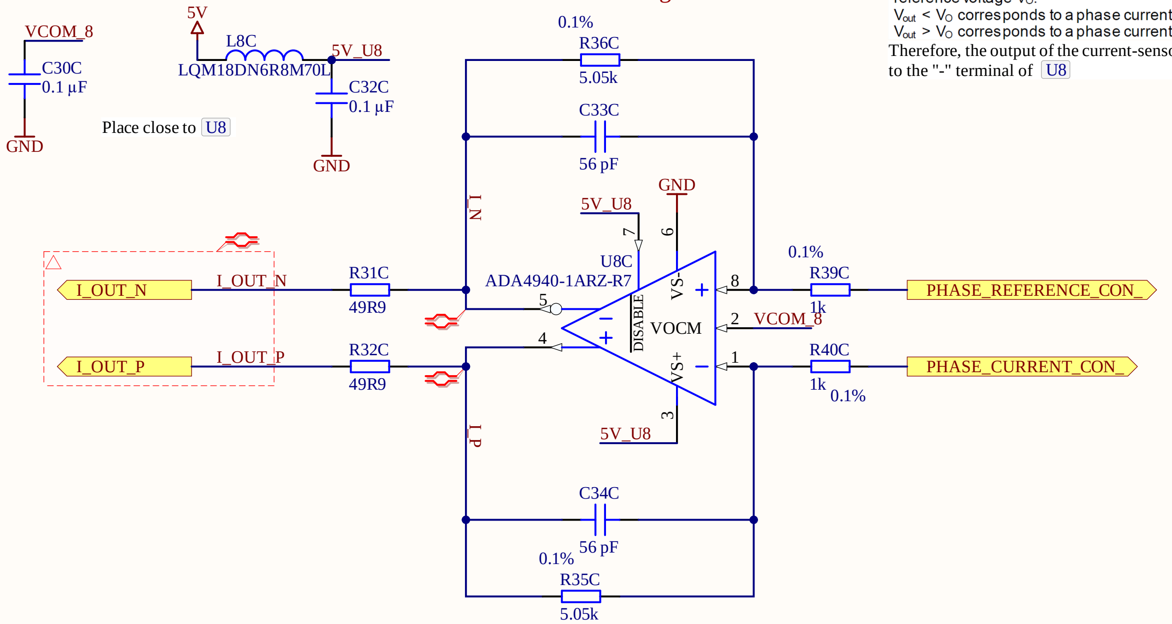
Project: UZ_PER_SKAI3_LV_48V_410A.PrjPCB		Date: 04.07.2024 Sheet 6.2 of 9

Phase-Current measurement single to differential



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Therefore, the output of the current-sensors of the inverter are connected to the "-" terminal of U8

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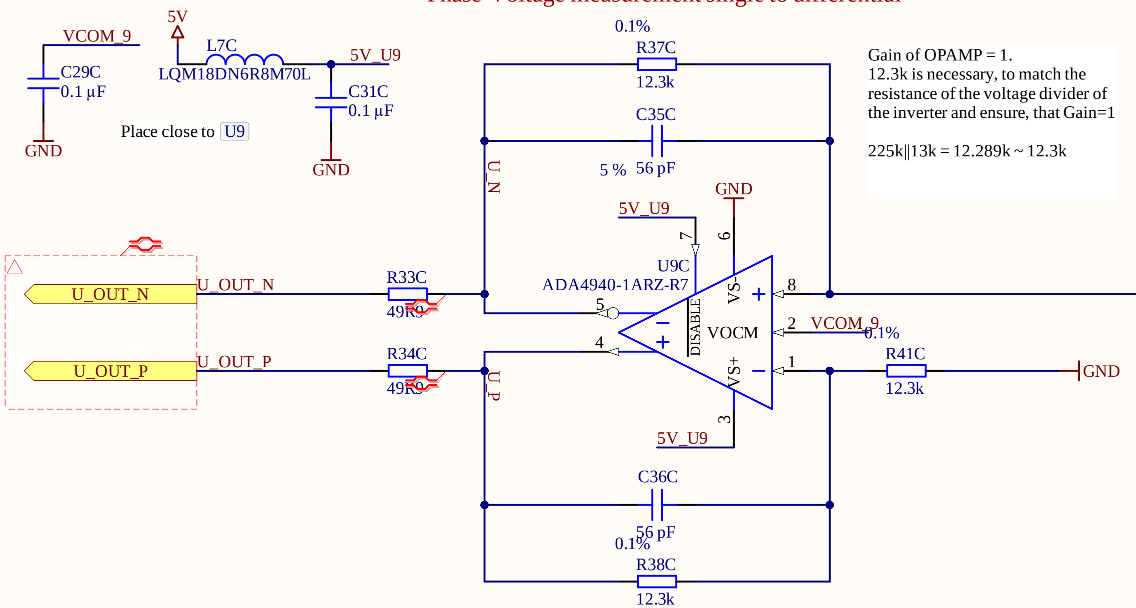
Phase-Voltage measurement single to differential



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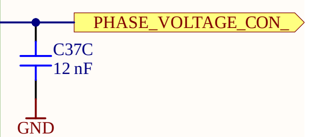

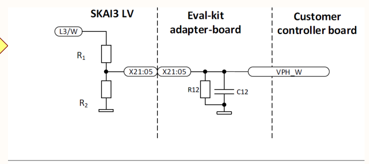

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Title Curr_Volt_Measurement.SchDoc	**UltraZohm** www.ultrazohm.com
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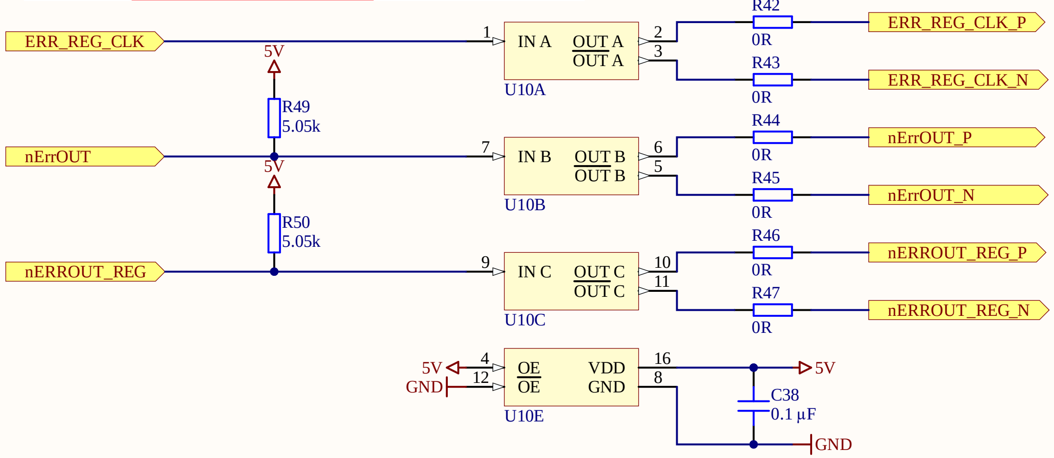
Project: UZ_PER_SKAI3_LV_48V_410A.PrjPCB	Date: 04.07.2024 Sheet 6.3 of 9

Pin No.	Pin Name	Description	Type	Electical characteristics
18	nErrOut	Indicates error state, low active Connect a pull-up resistor in range between 2 kΩ to 5.5 kΩ to positive controller I/O supply; of respective controller input pin chapter 9	digital output	Open drain; low active; R_{ON}<240 Ω
19	RESET	Gate driver reset signal, high active low: disable PWM, set nErrOut signal active high: release PWM rising edge: reset error state	digital input	low: 0 ...0.8 V high: 2.4 V ... Vs R_{IN} = 50 kΩ
20	nERROUT_REG	Gate driver error register output (alternating polarity in each readout cycle of the entire error register, starting low active). Connect a pull-up resistor in range between 2 kΩ to 5.5 kΩ to positive controller I/O supply of respective controller input pin; see chapter 9	digital output	Open drain; R_{ON}<240 Ω



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